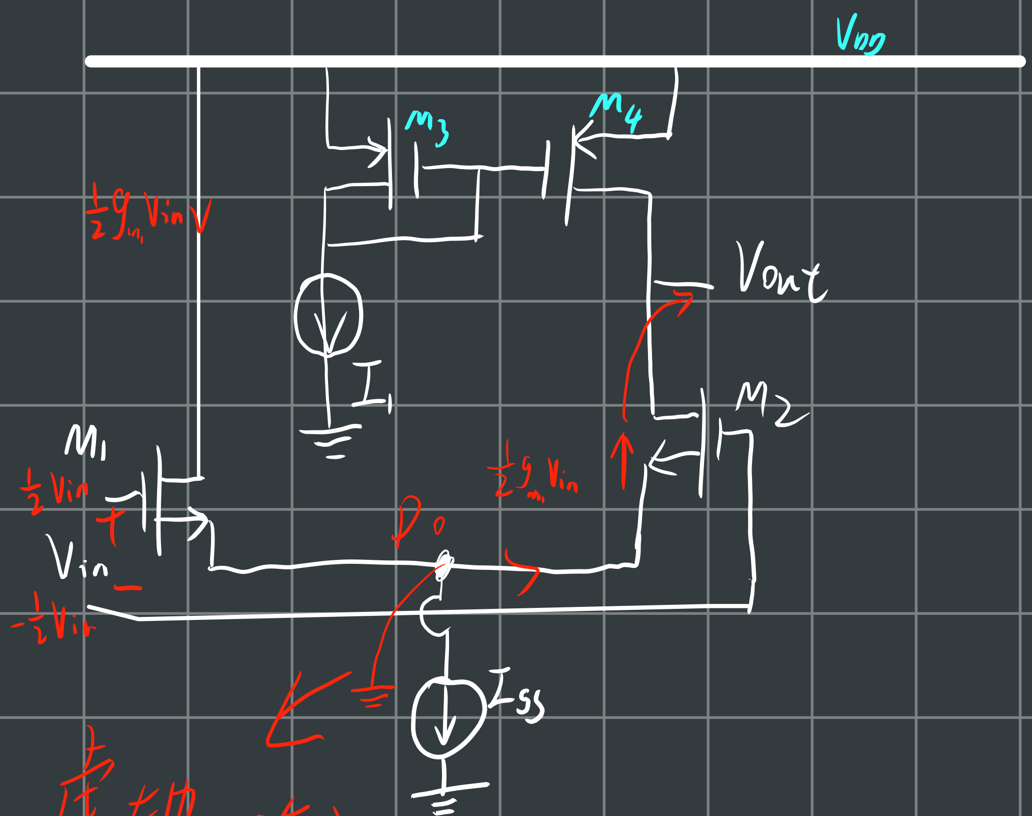
5.7 图中电路 $(W/L)_{1-4} = \frac{50}{0.5}$ 且 $I_{SS} = 2I_1 = 0.5mA$

(a) 计算小信号电压增益

(b) 如果输入共模电压是1.3V，确定最大输入输出电压摆幅



虚地，左边电压失效

(b) 在M_1、M_2饱和的情况下

电压摆幅为 $2 \times (V_{out\,max} - V_{out\,min})$

$$V_{out\,min} = V_{CM} - V_{GS2} + V_{OD2}$$
$$= V_{cm} - V_{th2}$$
$$= 0.6V$$

$$V_{out\,max} = V_{DD} - V_{OD4}$$
$$= V_{DD} - \sqrt{\frac{2I_{D4}}{K_4'(\frac{W}{L})_4 I_{D4}}}$$
$$= \ldots\ldots$$

(a) $A_V = \frac{V_{out}}{V_{in}} = \frac{V_{out}}{2 \times \frac{V_{in}}{2}}$

$$= \frac{1}{2} \times (-1) \boxed{\frac{V_{out}}{(-\frac{V_{in}}{2})}} = -\frac{1}{2} \boxed{(-g_{m2} r_{out})}$$

$$= +\frac{1}{2} g_{m2} r_{out}$$

$$g_{m1,2} = \sqrt{2K_1' (\frac{W}{L})_1 I_{D1}}$$
$$= 2.345\,mA$$

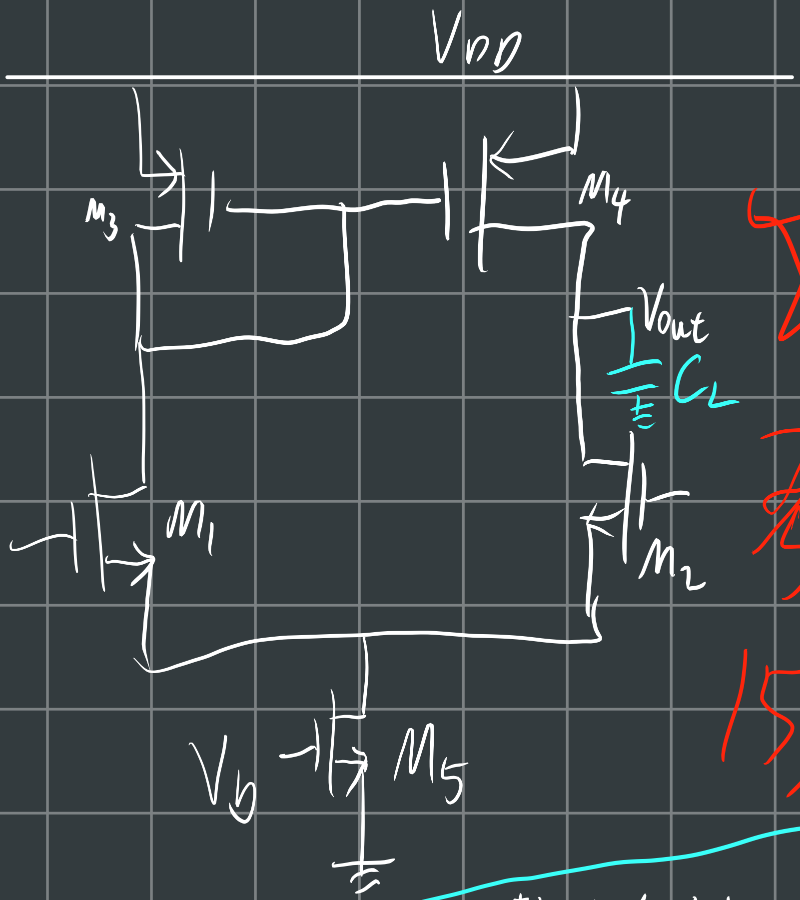
$$r_2 = \frac{1}{\frac{0.04}{V} \times 0.25mA} = \ldots$$

$$r_4 = \frac{1}{\frac{0.05}{V} \times 0.25mA} = \ldots$$

$$R_{out} = (2r_2) // r_4 = \ldots$$

求小信号，V_{DD}、V_b 虚地

求内阻，电流源开路



$V_{DD} = -V_{SS} = 2.5V$

$SR \geq 5 V/\mu s$, $C_L = 5pF$

$f_{-3dB} \geq 100 KHz$

$A_V = 100 V/V$

$-1.5 \leq ICMR \leq 2$

$P_{diss} \leq 0.5 mW$, $\lambda_n = 0.04V^{-1}$, $\lambda_p = 0.05V^{-1}$

考原图

15分+

→表达式只有一个未知量

从表达式简单的指标或电流入手

$$P_{diss} = I_5(V_{DD} - V_{SS}) \leq 0.5\ mW$$
$$I_5 \leq 0.1\ mA$$

① $SR = \frac{I_5}{C_L} = \frac{I_5}{5pF} \geq 5 V/\mu s$

$I_5 \geq 25\mu A$

② $\omega_{-3dB} = 2\pi f_{-3dB} = \frac{1}{R_{out} C_L} = \frac{1}{R_{out}\cdot 5pF}$

$\Rightarrow f_{-3dB} = \frac{1}{2\pi R_{out}\cdot 5pF} \geq 100kHz$

$\Rightarrow R_{out} \leq \frac{1}{2\pi\times 5pF\times 100kHz} = 318k\Omega$

又∵ $R_{out} = \frac{1}{(\lambda_2+\lambda_4)I_2} \leq 318k\Omega$

$\Rightarrow I_2 \geq \frac{1}{(\frac{0.04}{V}+\frac{0.05}{V})\times 318k\Omega} = 35\mu A$

③ $P_{diss} = 2V_{DD}I_5 = 5\times I_5 \leq 0.5mW$

∴ $I_5 \geq 100\mu A$

综合①②③: $35\mu A \leq I_5 \leq 100\mu A$ 取 $I_5 = 80\mu A$

④

$$|A_V| = g_{m2}(r_{o2} \parallel r_{o4})$$

$$= \sqrt{2 I_2 K_2' \left(\frac{W}{L}\right)_2} \times \frac{1}{\left(\frac{0.04}{V} + \frac{0.05}{V}\right) I_2}$$

$$= \sqrt{80\mu A \, \frac{110\mu A}{V^2} \left(\frac{W}{L}\right)_2} \times \frac{1}{\frac{0.09}{V} \times 40\mu A}$$

$$= 100\, V/V$$

$$\Rightarrow \left(\frac{W}{L}\right)_{2\text{-}3} = \frac{14.7\mu m}{1\mu m}$$

⑤

M_5 临界饱和

$$V_{in.min} = V_{SS} + V_{DS5} + V_{GS1} = -1.5$$

$$\Rightarrow -2.5V + \sqrt{\frac{2 I_5}{K'\left(\frac{W}{L}\right)_5}} + \left(\sqrt{\frac{2 I_1}{K'\left(\frac{W}{L}\right)_1}} + V_{th1}\right) = -1.5V$$

$$\Rightarrow -2.5V + \sqrt{\frac{2 \times 80\mu A}{\frac{110\mu A}{V^2} \times \left(\frac{W}{L}\right)_5}} + \left(\sqrt{\frac{2 \times 40\mu A}{\frac{110\mu A}{V^2} \times 14.7}} + 0.7V\right) = -1.5$$

$$\Rightarrow \left(\frac{W}{L}\right)_5 = \frac{23.9\mu m}{1\mu m}$$

⑥

$M_{1\text{-}2}$ 临界饱和

$$V_{DS1} \geq V_{GS1} - V_{th1}$$

$$V_{G1} \leq V_{D1} + V_{th1}$$

$$\Rightarrow V_{G1} \leq (V_{DD} - |V_{GS3}|) + V_{th1}$$

$$V_{in.max} = V_{DD} - |V_{GS3}| + V_{th1}$$

$$2V = 2.5V - \left(\sqrt{\frac{2 I_3}{K_3'\left(\frac{W}{L}\right)_3}} + V_{th3}\right) + V_{th1}$$

$$2V = 2.5V - \left(\sqrt{\frac{2 \times 40\mu A}{\frac{50\mu A}{V^2}\left(\frac{W}{L}\right)_3}} + 0.7V\right) + 0.7V$$

$$\Rightarrow \left(\frac{W}{L}\right)_{3\text{-}4} = \frac{6.4\mu m}{1\mu m}$$

⑦

$$V_b = ?$$